E-Commerce Sales Analysis

Observation – Power BI Assignment 2: DAX & Data Visualization

1. Project Overview

This project focused on analysing e-commerce sales data using Power BI by applying data modelling, DAX calculations, and interactive visualizations. The objective was to transform raw sales data into meaningful business insights that support decision-making and performance evaluation.

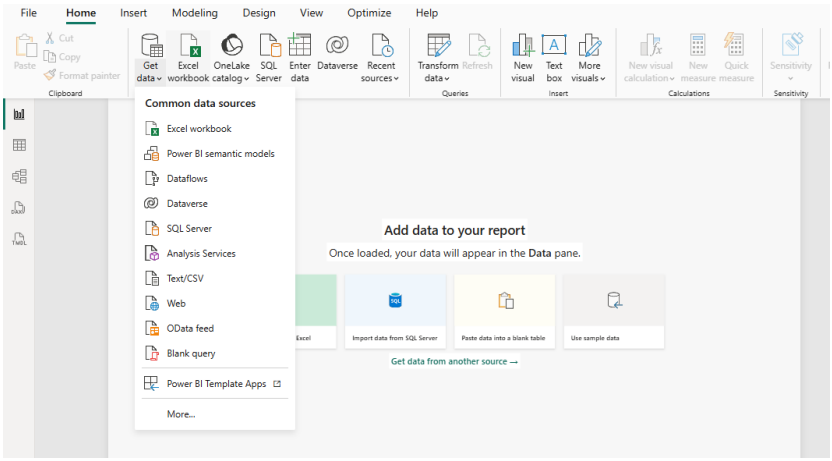
2. Data Sources Used

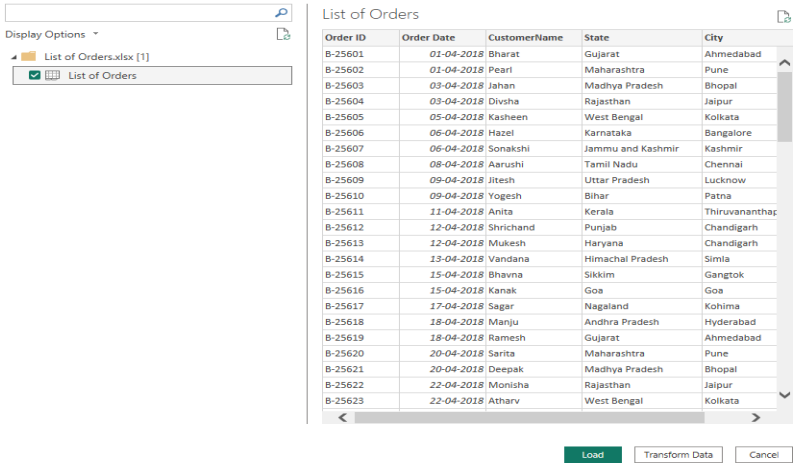
The analysis was performed using the following datasets:

★ List of Orders
★ Order Details
★ Sales target

Data Transformation Process:

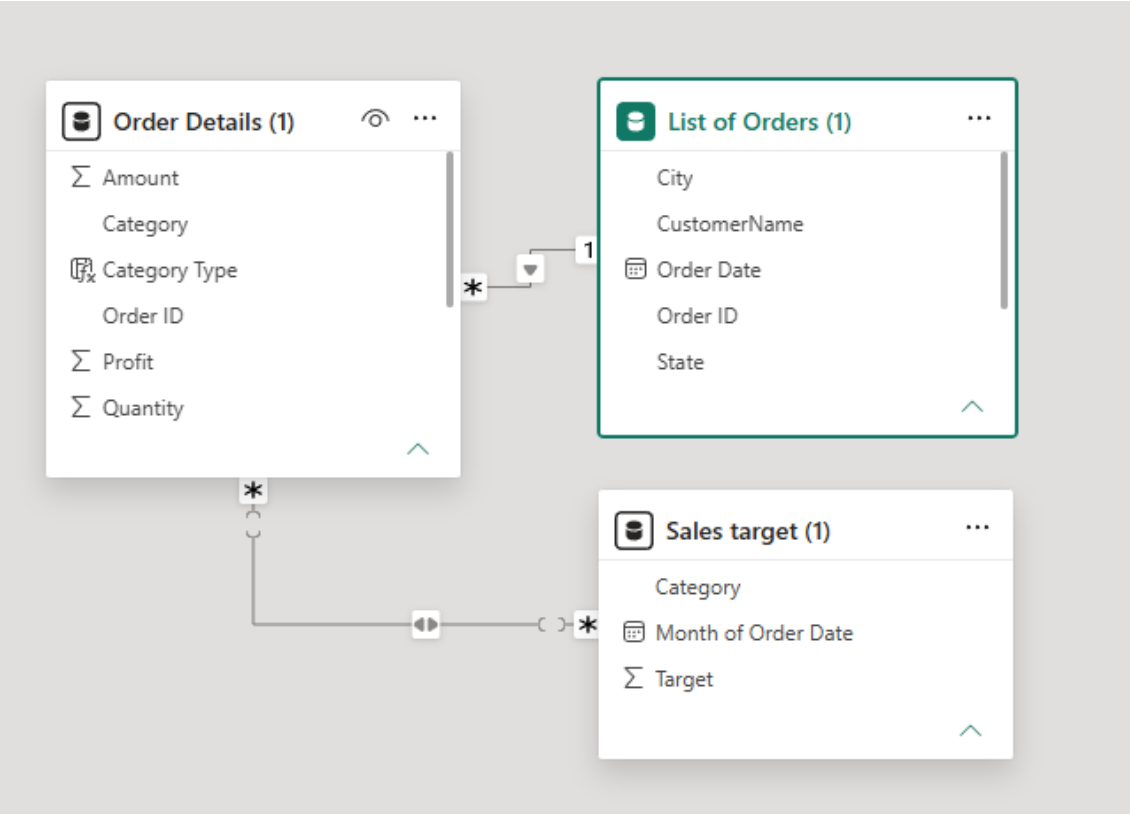
1. Imported the **List of Orders**, **Order Details**, and **Sales Target** datasets into Power BI





Data Modelling:

- Verified the Data Set and No duplicates and missing values found.
- Establish a relationship between the “List of Orders” and “Order Details” tables using the ‘Order ID’ column.



DAX Calculated Columns

Category Type

A calculated column was created by combining Category and Sub-Category fields to provide a more detailed product classification.

```
1 Category Type = 'Order Details (1)'[Category]&"-"&'Order Details (1)'[Sub-Category]
```

Category	Sub-Category	Category Type	Rev
Clothing	Saree	Clothing-Saree	
Clothing	Saree	Clothing-Saree	
Clothing	Saree	Clothing-Saree	
Clothing	Saree	Clothing-Saree	
Clothing	Saree	Clothing-Saree	
Clothing	Saree	Clothing-Saree	
Clothing	Saree	Clothing-Saree	
Clothing	Saree	Clothing-Saree	
Clothing	Saree	Clothing-Saree	
Clothing	Saree	Clothing-Saree	
Clothing	Saree	Clothing-Saree	
Clothing	Saree	Clothing-Saree	
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Clothing	Saree	Clothing-Saree	
Clothing	Saree	Clothing-Saree	
Clothing	Saree	Clothing-Saree	
Clothing	Saree	Clothing-Saree	

Revenue per Order

A new column was created to calculate revenue generated per order using the formula

```
1  Revenue per Order = 'Order Details (1)'[Amount]*'Order Details (1)'[Quantity]
```

Revenue = Amount × Quantity

```
ails (1)'[Quantity]
```

Category Type	Revenue per Order
Clothing-Saree	1683
Clothing-Saree	952
Clothing-Saree	579
Clothing-Saree	1413
Clothing-Saree	525
Clothing-Saree	100
Clothing-Saree	129
Clothing-Saree	320
Clothing-Saree	14427
Clothing-Saree	2824
Clothing-Saree	1602
Clothing-Saree	596
Clothing-Saree	3175
Clothing-Saree	96
Clothing-Saree	2844
Clothing-Saree	1146
Clothing-Saree	3185
Clothing-Saree	488
Clothing-Saree	40
Clothing-Saree	168

Sales Category

Orders were categorized as **Above Average** or **Below Average** based on their sales amount relative to the average sales value.

```
1 Sales Category = IF('Order Details (1)'[Amount]>750,"Above Average",IF('Order Details (1)'[Amount]<750,"Below Average"))
```

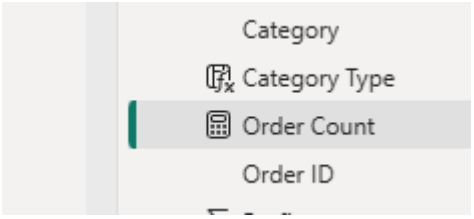
Revenue per Order	Sales Category
1683	Below Average
952	Below Average
579	Below Average
1413	Below Average
525	Below Average
100	Below Average
129	Below Average
320	Below Average
14427	Above Average
2824	Below Average
1602	Below Average
596	Below Average
3175	Below Average
96	Below Average
2844	Below Average
1146	Below Average
3185	Below Average
488	Below Average

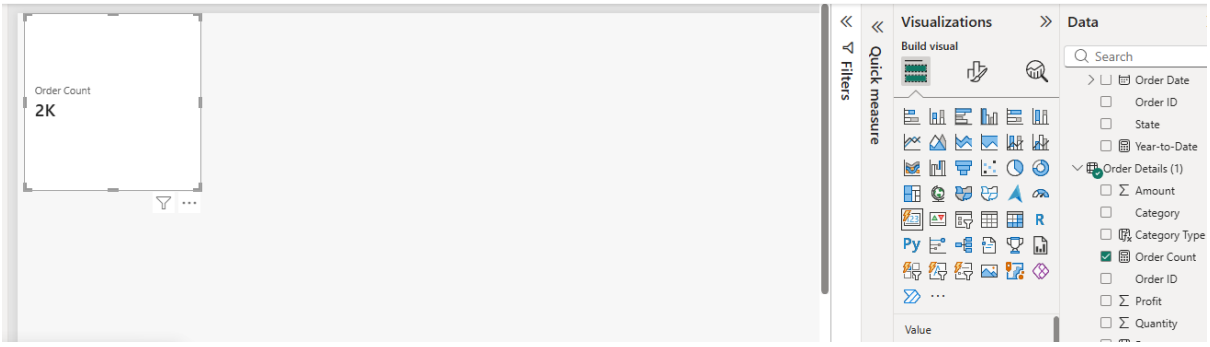
DAX Measures

Order Count

A measure was developed to calculate the total number of orders processed in the dataset.

```
1 Order Count = COUNTROWS('Order Details (1)')
```

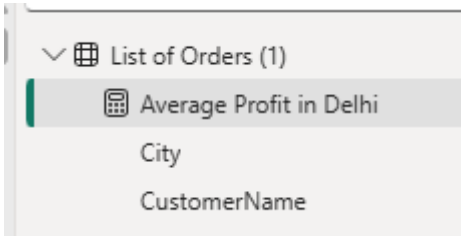


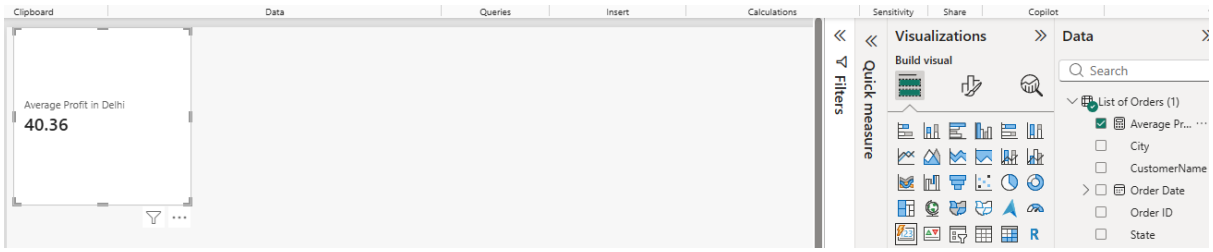


Average Profit in Delhi

A DAX measure was created to determine the average profit generated from orders placed in Delhi.

```
1  Average Profit in Delhi = CALCULATE(AVERAGE('Order Details (1)'[Profit]),'List of Orders (1)'[State]="Delhi")
```

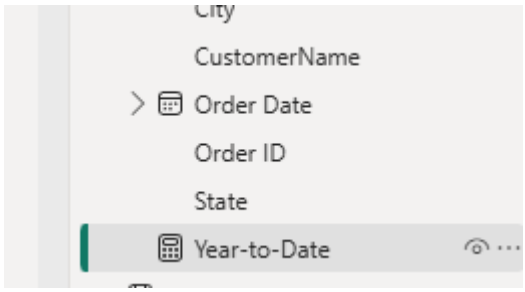


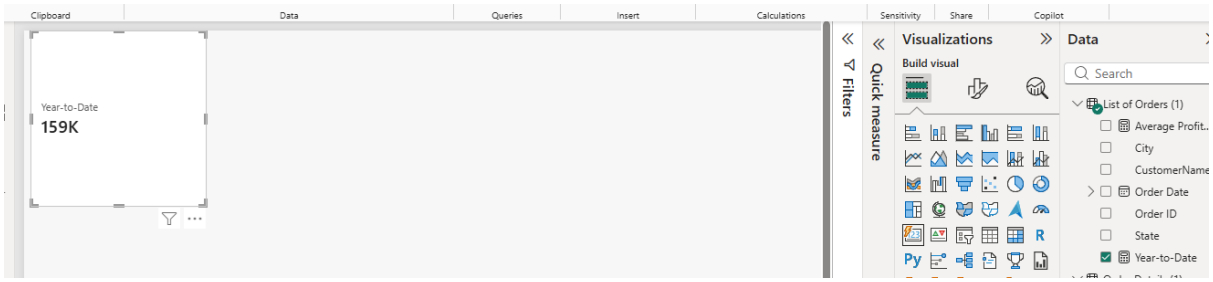


Year-to-Date (YTD) Sales

A cumulative sales measure was developed to track total sales from the beginning of the year up to each transaction date.

```
1  Year-to-Date = TOTALYTD(SUM('Order Details (1)'[Amount]),'List of Orders (1)'[Order Date])
```

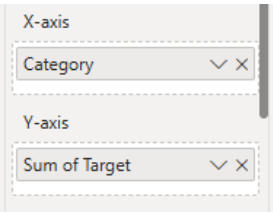


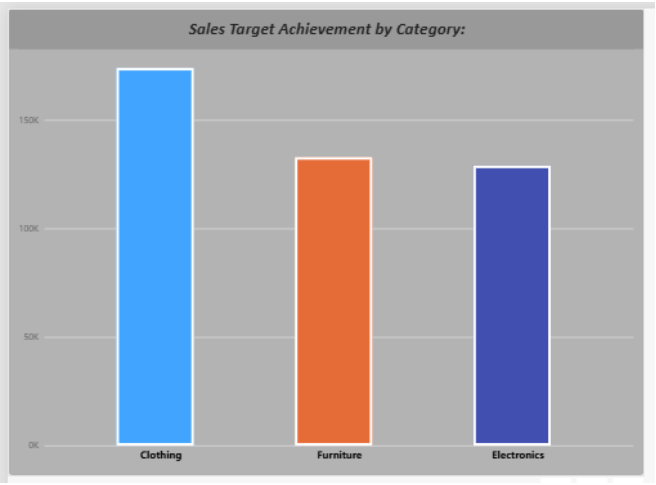


Data Visualizations

Sales Target Achievement by Category

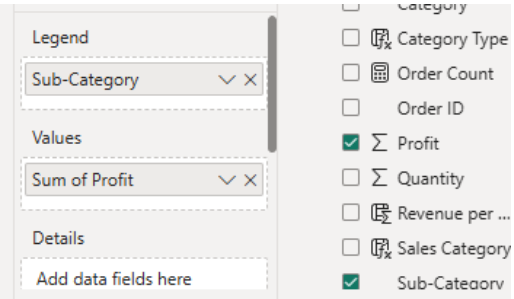
A clustered column chart was used to compare actual sales with target sales across different product categories.

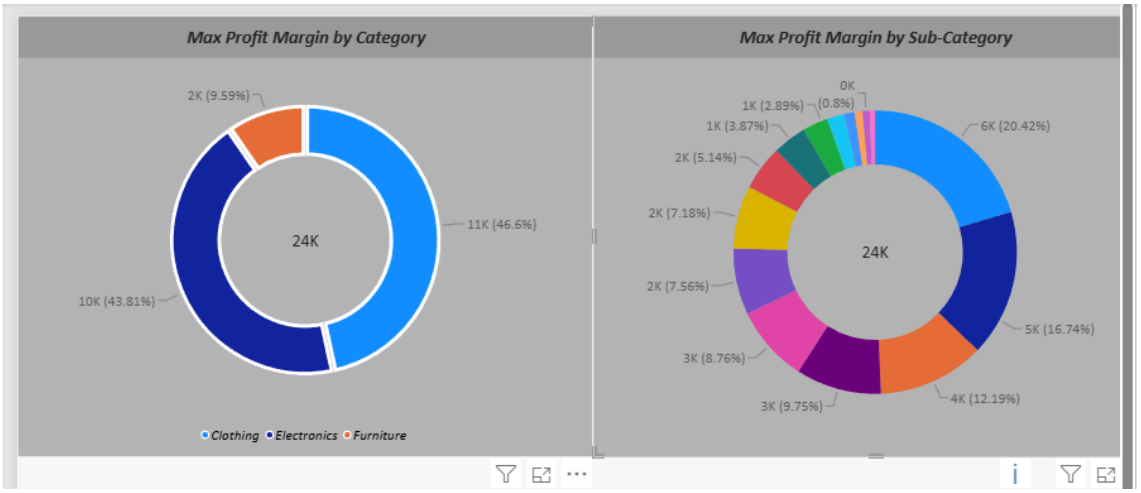




Maximum Profit Margin by Sub-Category

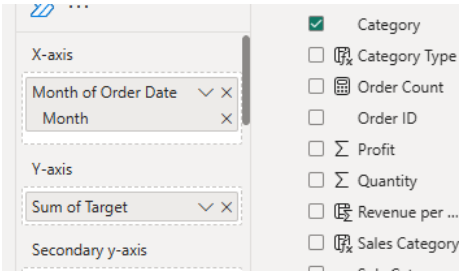
A donut chart was created to identify sub-categories contributing the highest profit margins.

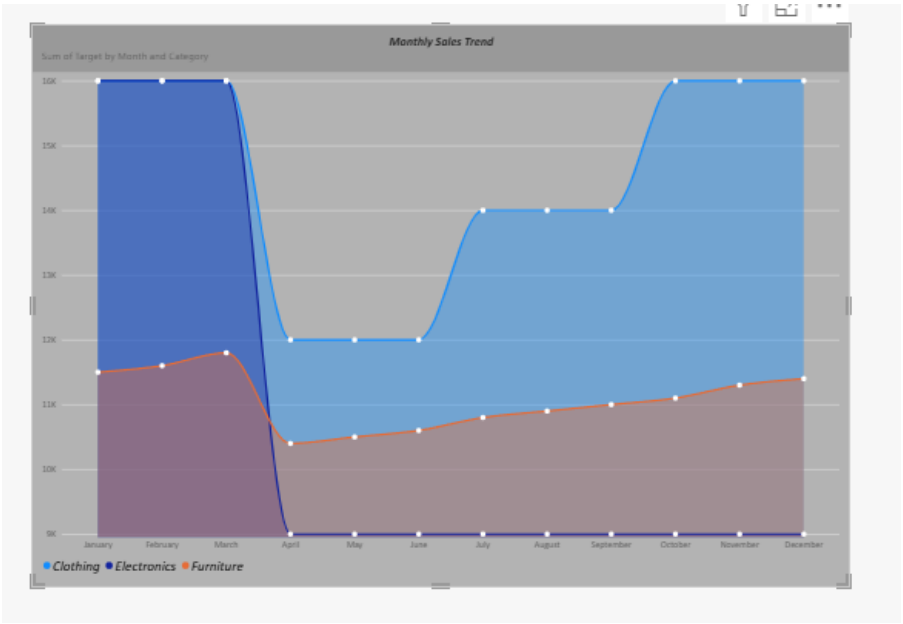




Monthly Sales Trend

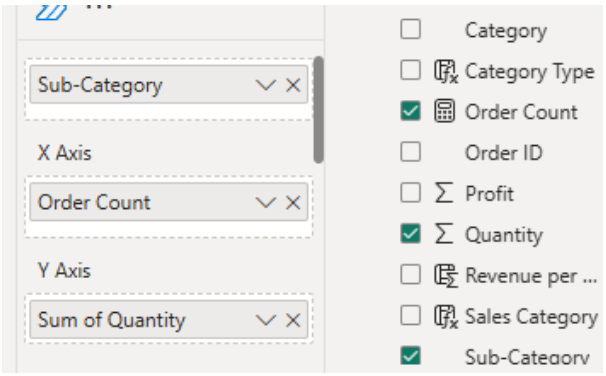
A line chart displayed monthly sales performance and highlighted overall sales trends over time.





Profit vs Quantity Analysis

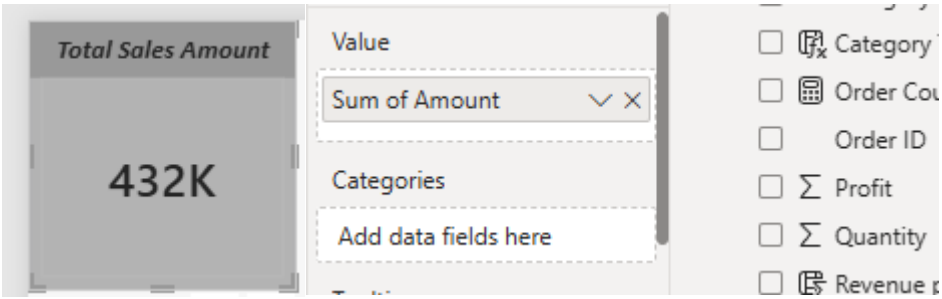
A scatter chart was used to examine the relationship between profit earned and quantity sold for different product sub-categories.

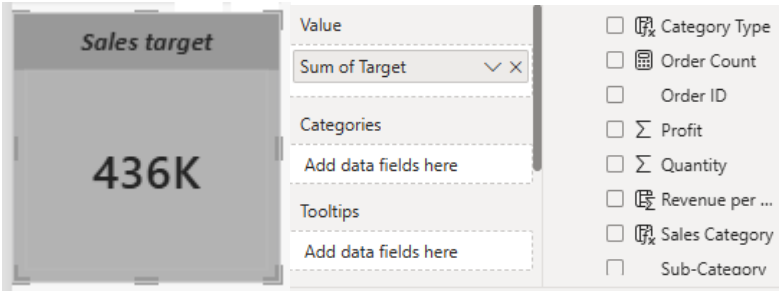




Total Sales and Target Comparison

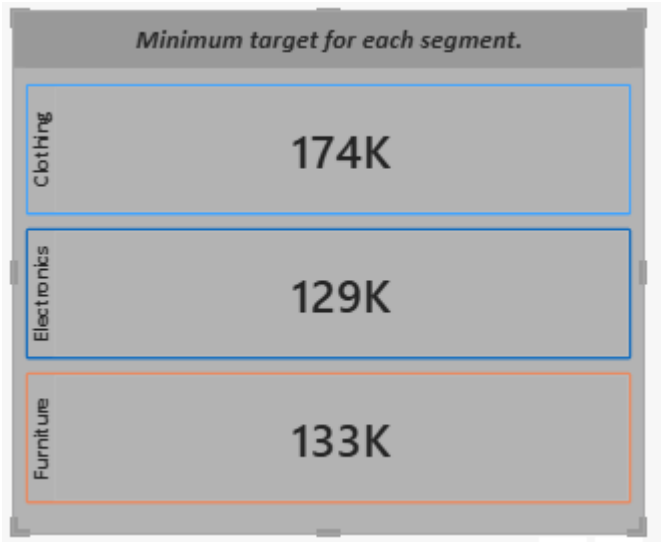
Card visuals were designed to display total sales amount and target values for quick performance monitoring.

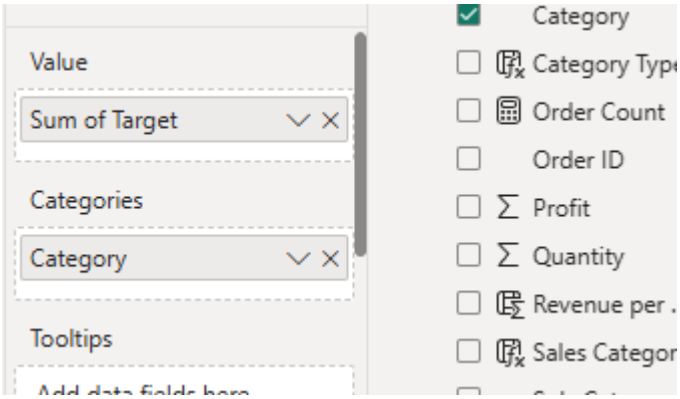




Minimum Target Analysis

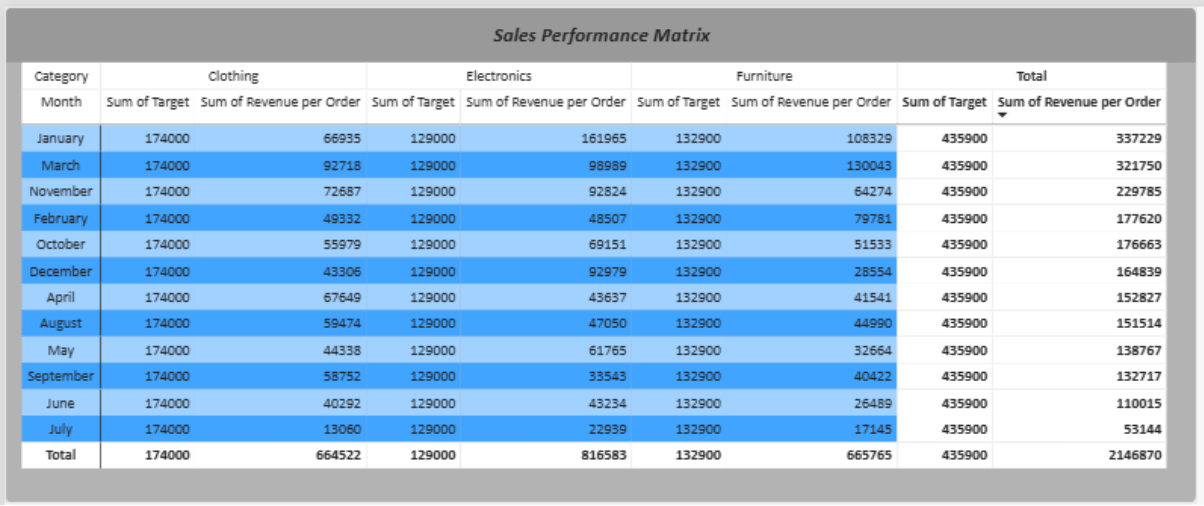
A multi-row card was used to present the minimum sales target for each customer segment.





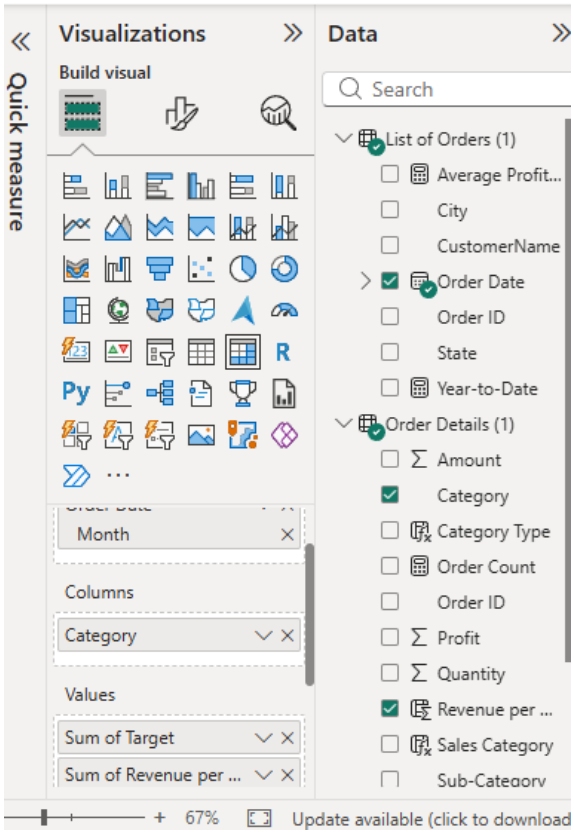
Sales Performance Matrix

A matrix visualization provided a detailed comparison of actual sales against target sales by category and month.



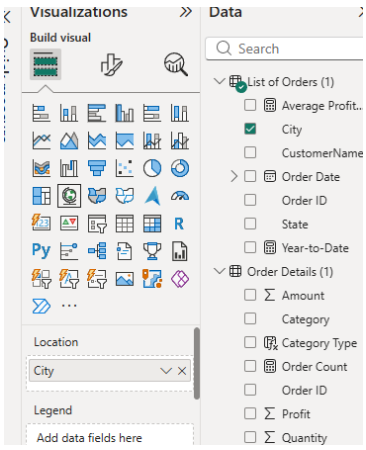
Sales Performance Matrix

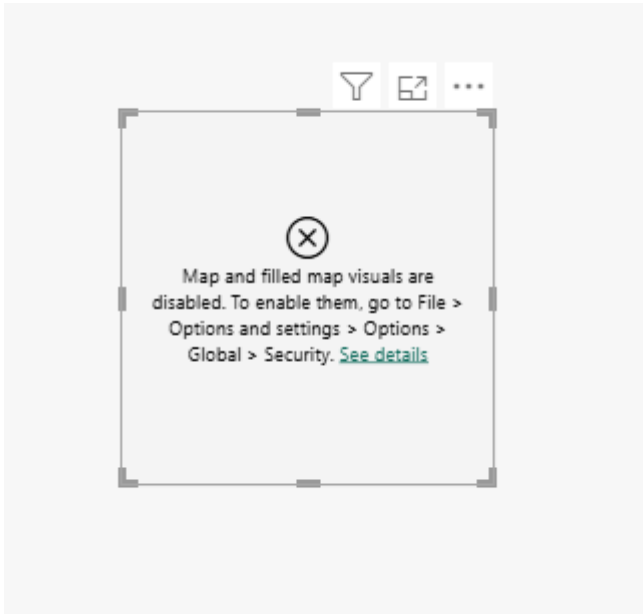
Category	Clothing		Electronics		Furniture		Total	
Month	Sum of Target	Sum of Revenue per Order	Sum of Target	Sum of Revenue per Order	Sum of Target	Sum of Revenue per Order	Sum of Target	Sum of Revenue per Order
January	174000	66935	129000	161965	132900	108329	435900	337229
March	174000	92718	129000	98989	132900	130043	435900	321750
November	174000	72687	129000	92824	132900	64274	435900	229785
February	174000	49332	129000	48507	132900	79781	435900	177620
October	174000	55979	129000	69151	132900	51533	435900	176663
December	174000	43306	129000	92979	132900	28554	435900	164839
April	174000	67649	129000	43637	132900	41541	435900	152827
August	174000	59474	129000	47050	132900	44990	435900	151514
May	174000	44338	129000	61765	132900	32664	435900	138767
September	174000	58752	129000	33543	132900	40422	435900	132717
June	174000	40292	129000	43234	132900	26489	435900	110015
July	174000	13060	129000	22939	132900	17145	435900	53144
Total	**174000**	**664522**	**129000**	**816583**	**132900**	**665765**	**435900**	**2146870**



Geographic Sales Analysis

A map visualization was developed to analyse city-wise sales performance and identify regional sales patterns.

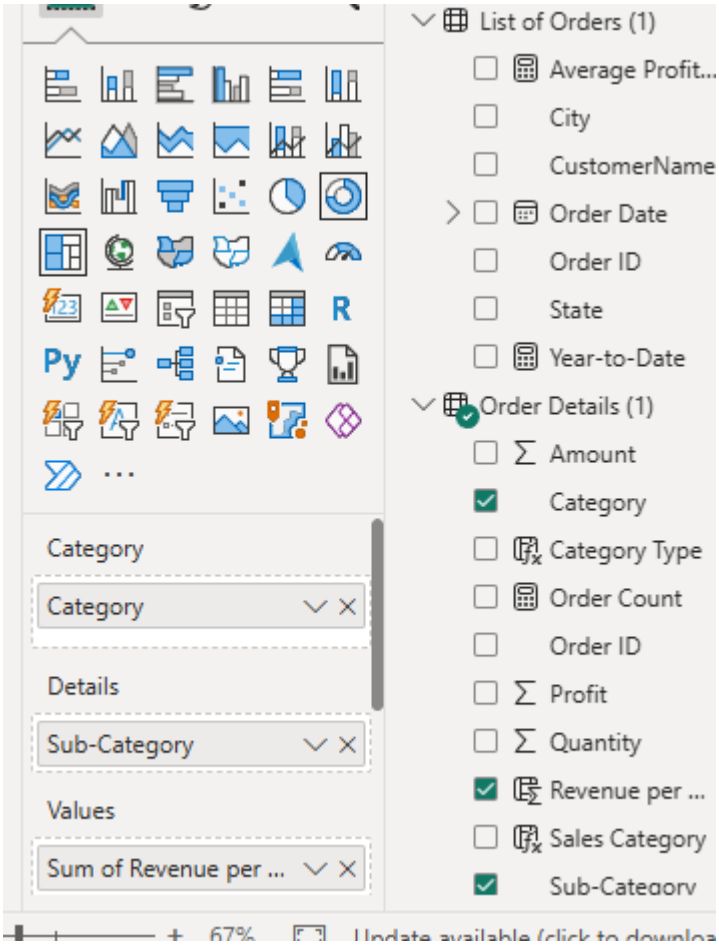


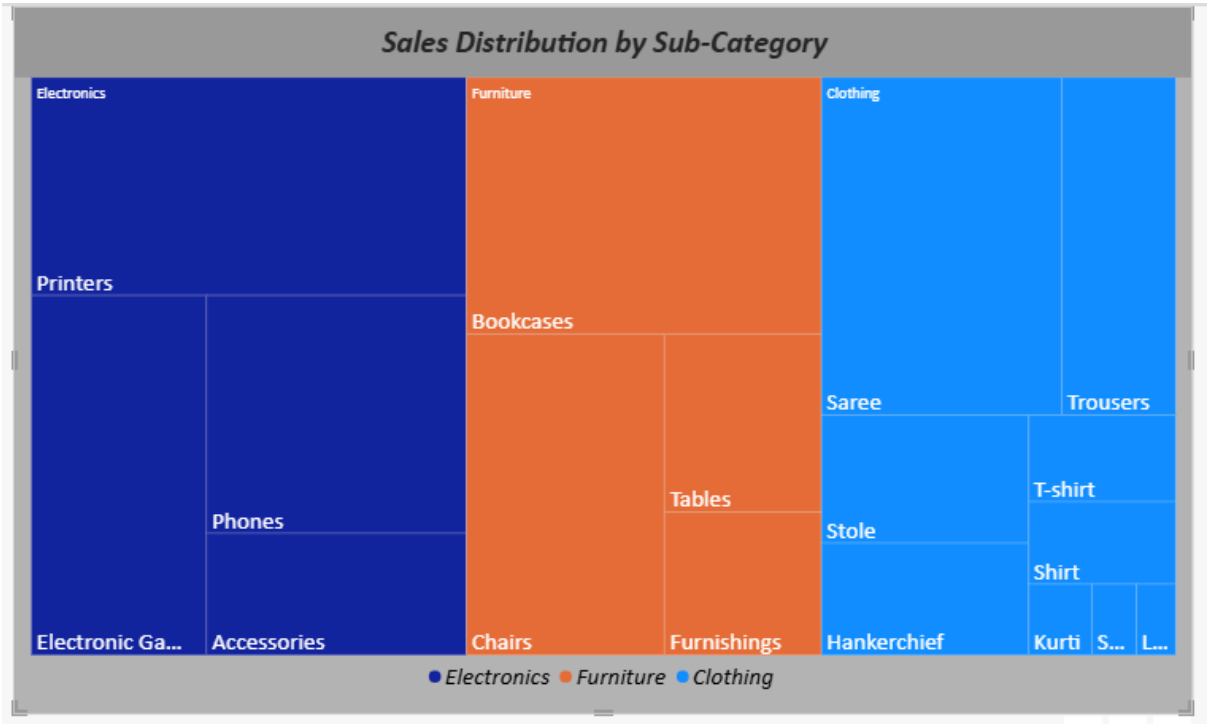


Due to an Error I came up with this Result, also adjusted the setting as mentioned but same results.

Sales Distribution by Sub-Category

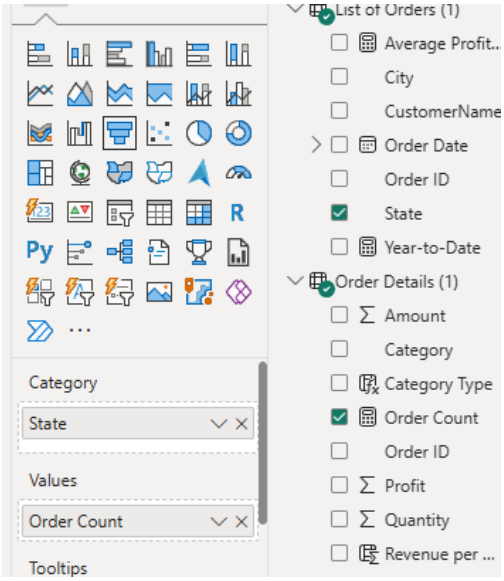
A tree map visual displayed the contribution of different sub-categories to total sales.

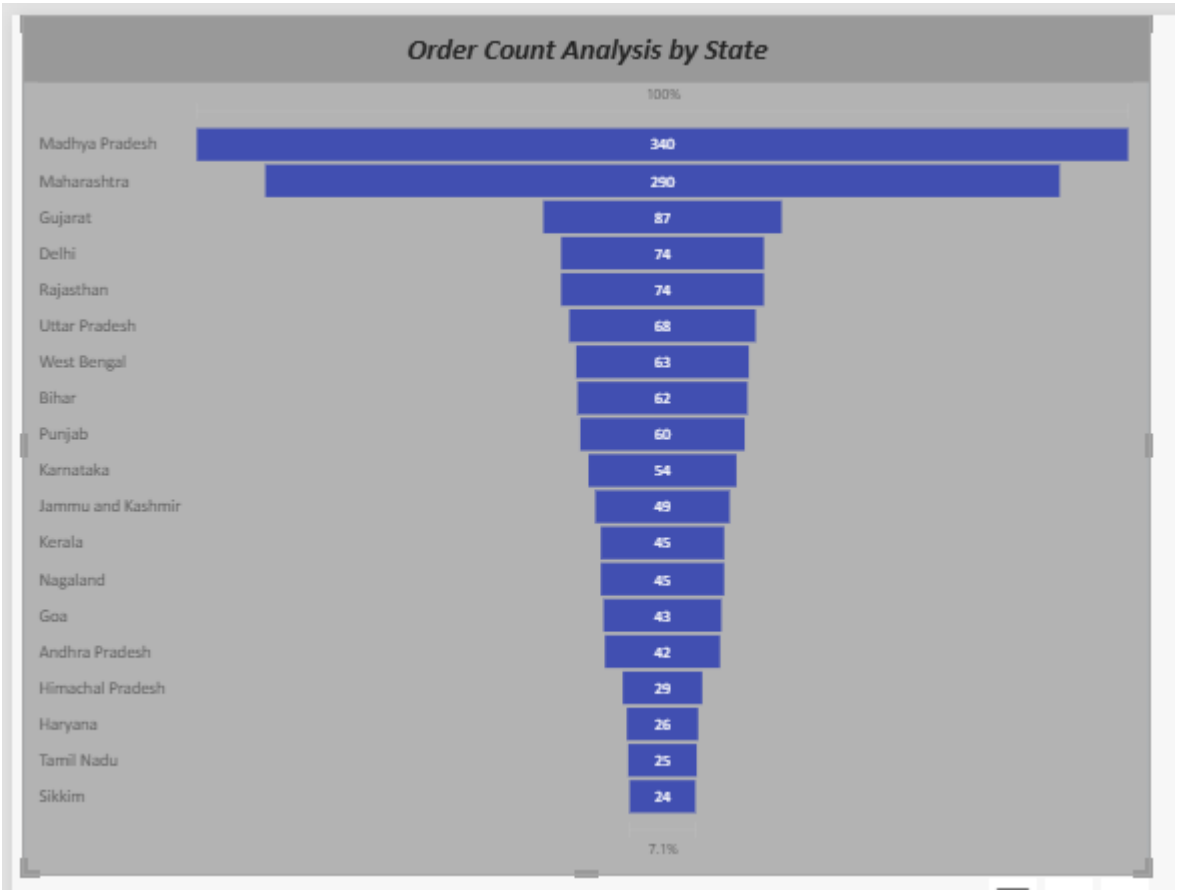




Order Count Analysis by State

A funnel chart illustrated the distribution of order counts across various states.





Conclusion

This project successfully demonstrated the use of Power BI for data modelling, DAX calculations and visualization development. The dashboard transformed raw e-commerce data into meaningful insights, enabling users to monitor sales performance, analyse profitability, track targets, and make data-driven business decisions effectively.

